describe this insight. Heisenberg's uncertainty paper paid off as he had hoped: the University of Leipzig again offered him a tenured professorship. He accepted, and in June 1927, Heisenberg, at twenty-five, became the youngest tenured professor in all of Germany.

Meanwhile, Bohr found that Heisenberg's uncertainty principle meshed well with his own new ideas about the true nature of the quantum world, which he called "complementarity." In typical Bohr fashion, his paper on complementarity became bogged down in a series of drafts filled with sentences that refused to end. But that September, Bohr ran out of time for rewrites. The International Physics Conference was meeting on the shores of alpine Lake Como, in northern Italy, and Bohr was scheduled to give the keynote address. Frantically revising his prepared statements up through the day of his talk, Bohr took the stage, speaking softly and haltingly.

Bohr started from the idea that "our usual description of physical phenomena is based entirely on the idea that the phenomena concerned may be observed without disturbing them appreciably." But, as Heisenberg's uncertainty principle made clear, "any observation of atomic phenomena will involve an interaction with the agency of observation not to be neglected." Therefore, Bohr continued, "an independent reality in the ordinary physical sense can neither be ascribed to the phenomena nor to the agencies of observation." In other words, one could not ask what was really happening inside of an atom when nobody looked—according to Bohr, the quantum world could only be considered real in conjunction with some kind of measurement apparatus to study that world. And the behavior of the objects in that world, as indicated by such an apparatus, would be best described as either particles or waves, but never both simultaneously. These descriptions are contradictory—a particle has a definite location, which waves don't; waves have frequencies and wavelengths, which particles don't—yet Bohr claimed that this "inevitable dilemma" was not a problem for quantum physics. "We are not dealing with contradictory but with complementary pictures of the phenomena," claimed Bohr, which are "indispensable for a description of experience."

This "wave-particle duality" shows up in all quantum phenomena. For example, in an old cathode-ray-tube TV, electrons shoot from the back of the TV toward the phosphorescent screen at the front of the TV, which lights up when an electron hits it. When an electron is shot out into the tube,